DRAFT VERSION JULY 23, 2026
Typeset using LATEX **twocolumn** style in AASTeX631

The $z \simeq 4$–6 Massive-Galaxy Abundance is Consistent with IllustrisTNG Once the Stellar-Mass Systematic Budget and Aperture Basis are Accounted For

NEBULAMIND LAB (AUTONOMOUS RUN)[1]

[1] *Descriptive, machine-generated draft — not a validated measurement.*

ABSTRACT

The reported over-abundance of massive galaxies at $z > 4$ relative to ΛCDM hydrodynamical simulations ("too massive, too early") has been read as possible evidence for new physics. We test this by confronting IllustrisTNG (TNG100-1) cumulative number densities $n(> M_\star, z)$ against JWST-era observations on a like-for-like stellar-mass basis, and by benchmarking the observed abundance against the ΛCDM baryon-conversion ceiling. Because the observed SED masses are total-galaxy quantities, we place the simulation on the same footing: TNG100-1 $M_\star$ in this work is the within-2 $R_{1/2}$ aperture, and the aperture-to-total correction is +0.13 dex, giving $n(> 10^{10.5}\,M_\odot) = 1.47 \times 10^{-5}$ Mpc^{-3} at $z = 5$ (from 20 subhalos in the $(110.7\,\mathrm{Mpc})^3$ box). At $z \simeq 5$ the observed $n(> 10^{10.5}\,M_\odot)$ (Weibel et al. 2024, $\approx 3 \times 10^{-5}$ Mpc^{-3}) then exceeds TNG by a factor ≈ 2.04 (0.31 dex); given the steep massive-end slope of the simulated stellar mass function ($d \log n/d \log M_\star \approx -1.58$), this excess is erased by a downward stellar-mass shift of only ≈ 0.20 dex. We replace the loosely quoted "∼ 1 dex" budget with an itemized, six-axis stellar-mass systematic ledger whose realistic (independent-quadrature) value is 0.55 dex — 0.46 dex if the contested top-heavy-IMF term is excluded, and 1.30 dex only in the fully correlated worst case. The required 0.20 dex shift is ∼ 0.4× the committed budget and is covered even without invoking a top-heavy IMF; the $z \simeq 5$ consistency is therefore robust and IMF-independent, subject to an irreducible ±0.10 dex Poisson floor from the ≈ 15–20 simulated galaxies that set the anchor. Crucially, the *unshifted* observed abundance already implies a baryon-conversion efficiency $\epsilon = M_\star/(f_b\,M_{\rm halo}) \approx 0.20$ at the abundance-matched halo mass $M_{\rm halo} = 1.0 \times 10^{12}\,M_\odot$ ($f_b = 0.157$) — i.e. the fiducial ΛCDM star-formation efficiency, far below the $\epsilon \le 1$ hard ceiling. The $z \simeq 5$ offset is thus not a ΛCDM stress test at all but a mismatch against TNG's *specific* feedback/SMF calibration; ΛCDM feasibility would be breached only if the true masses were ≈ +0.70 dex *higher* than reported (opposite in sign, and 2.5× larger than any plausible downward budget). The larger apparent excess at $z \simeq 7$–9 (Labbé et al. 2023 candidates, ≈ 13.6×, 1.13 dex) requires ≈ 0.72 dex at the same slope — which *exceeds* the committed 0.55 dex budget — and rests on unconfirmed photometric masses; we therefore label it outside the realistic budget and marginal, not consistent. We flag as a distinct, harder residual the spectroscopically confirmed quiescent galaxies at $z > 6$, whose ∼ 2 dex excess is not dissolved by these systematics. This is a descriptive confrontation of simulation predictions with observations on a matched stellar-mass basis; it is not a validated measurement.

1. INTRODUCTION

Early JWST photometry reported an unexpected abundance of luminous, apparently very massive galaxies at $z > 6$, implying baryon-to-stellar conversion efficiencies approaching or exceeding unity when compared to the ΛCDM halo mass function (Labbé et al. 2023). Taken at face value, such objects challenge standard structure formation. However, the inferred stellar masses are model-dependent quantities, and a growing body of spectroscopic and statistical work attributes much of the apparent tension to systematics rather than new physics. Here we ask a single, falsifiable question: *does the observed massive-galaxy abundance exceed the IllustrisTNG prediction by more than the stellar-mass systematic budget?*—and we separate that question, which probes TNG's calibration, from the distinct question of whether the abundance is physically possible in ΛCDM at all.

2. DATA AND METHOD

We take cumulative comoving number densities $n(> M_\star, z)$ from the public IllustrisTNG (TNG100-1) subhalo catalogs (Nelson et al. 2019), and compare to two observed anchors of $n(> 10^{10.5}\,M_\odot)$: Weibel et al. (2024) at $z \simeq 5$ (Weibel et al. 2024) and Labbé et al. (2023) candidates at $z \simeq 7$–9 (Labbé et al. 2023). The comparison is non-circular by construction: TNG masses are predictions, the observations are independent. TNG100-1 stellar masses in this work are the within-2×stellar-half-mass-radius aperture (`SubhaloMassInRadType`); we verify this reproduces the previously quoted $n(>10^{10.5}, z{=}5) = 1.1\times10^{-5}$ Mpc^{-3} (15 subhalos in the $(110.7\,\mathrm{Mpc})^3 = 1.357\times10^6$ Mpc^3 box), whereas the total gravitationally-bound stellar mass (`SubhaloMassType`) gives 1.47×10^{-5} Mpc^{-3} (20 subhalos). The observed SED masses are total-galaxy, so we place TNG on the same total-mass footing, an aperture correction of +0.13 dex (median over $\log M_\star > 10.3$ at $z = 5$; +0.12 dex at $z = 6$, so redshift-stable). We quantify the tension as the downward stellar-mass shift Δ required to bring the observed $n(>10^{10.5})$ into agreement with TNG, using the simulated massive-end slope $d\log n/d\log M_\star$ measured between $\log M_\star = 10.0$ and 10.5.

3. RESULTS

With TNG on the like-for-like total-mass footing, at $z = 5$ TNG gives $n(>10^{10.5}) = 1.47\times10^{-5}$ Mpc^{-3} ($N = 20$) and the observed value is 3×10^{-5}, a factor 2.04 (0.31 dex). With a local slope $d\log n/d\log M_\star \approx -1.58$, the excess is erased by $\Delta \approx 0.20$ dex. (On the raw $2\,R_{1/2}$ aperture the corresponding numbers are 1.11×10^{-5} Mpc^{-3}, 2.7×, 0.43 dex, and $\Delta \approx 0.28$ dex; the aperture fix therefore *shrinks* the required shift.) At $z \simeq 7$–9 the apparent excess is $\approx 13.6\times$ (1.13 dex) and requires $\Delta \approx 0.72$ dex at the same slope. The $z = 5$ anchor is Poisson-limited: it rests on 15 subhalos ($2\,R_{1/2}$; 20 on total masses) above $10^{10.5}$ in the box, a fractional error of ±26% (±0.10 dex), so both the 0.31 dex excess and the 0.20 dex required shift carry an irreducible ±0.10 dex cosmic-variance/Poisson floor, and the confrontation is one of tens of simulated massive galaxies against a single observational data point.

3.1. *An itemized stellar-mass systematic budget*

The claim that the required shift lies "within the budget" is only as good as the budget. We therefore replace the single "∼ 1 dex" figure — which is the code-to-code SED spread, and already *contains* the IMF, SPS and SFH drivers it is often added to — with a decomposition into independent physical axes (Table 1), each driving a downward revision of $M_\star$ for $z \approx 4$–6 massive JWST galaxies.

Combining these independently in quadrature gives a realistic committed budget of 0.55 dex; dropping the contested top-heavy-IMF term (#1) gives 0.46 dex; the fully correlated worst case (linear sum) is 1.30 dex — this last is the "∼ 1 dex" figure the earlier literature quotes, and it is an *upper* bound, not the realistic budget. We caution that terms #2–#4 are three manifestations of the same SED-fitting degeneracy and #6 is partly derived from the #1–#4 mass scatter, so treating all six as strictly independent mildly inflates the quadrature; a hostile accounting lands the committed budget in the range 0.46–0.55 dex, which we adopt rather than a single clean number.

Against this budget: the $z \simeq 5$ requirement of 0.20 dex is $\approx 0.4\times$ the committed budget and is covered even by the IMF-excluded 0.46 dex — so the $z \simeq 5$ consistency does *not* depend on a top-heavy IMF and is robust (within the ±0.10 dex Poisson floor above). The $z \simeq 7$–9 requirement of 0.72 dex *exceeds* the committed 0.55 dex quadrature budget and is met only under the fully correlated worst case; it is therefore marginal and photometric, and we group it with the quiescent excess as outside the realistic budget rather than with the secure $z \simeq 5$ result. The spectroscopic quiescent $z > 6$ excess ($\gtrsim 1.4$ dex required) exceeds even the 1.30 dex linear worst case and is genuinely outside budget.

3.2. *Redshift bracketing of the TNG anchor*

Because TNG evolves steeply at the massive end, a single "$z \simeq 5$–6" label is unsafe: TNG100-1 cumulative number densities above $10^{10.5}$ are $n(\mathrm{total}) = 1.47\times10^{-5}$ Mpc^{-3} at $z = 5$ ($N = 20$) and 2.95×10^{-6} Mpc^{-3} at $z = 6$ ($N = 4$), a factor 5.0 (0.70 dex) over $\Delta z = 1$ (the $2\,R_{1/2}$ $z = 6$ value, 7.4×10^{-7} Mpc^{-3}, rests on a single object and is not used). We therefore pin the comparison to the observed sample's median redshift rather than across a "5–6" range. The Weibel et al. (2024) anchor is the rest-optical-selected SMF at $z \sim 5$, so we adopt $z = 5$, where the required shift is 0.20 dex (total-matched). We note as an explicit caveat that if the effective comparison redshift is instead $z \simeq 5.5$, TNG interpolates to $\approx 6.6\times10^{-6}$ Mpc^{-3} and the required shift rises to ≈ 0.42 dex — still within the 0.46–0.55 dex committed budget, but with the margin nearly gone.

Table 2 generalizes the headline point: the downward stellar-mass shift required to erase the excess stays within the committed budget across the plausible range of observed excess (2–20×) and simulated massive-end slope (−1.4 to −2.0). Only a genuine ∼ 2 dex excess — the regime of the spectroscopic quiescent galaxies — would require $\gtrsim 1.4$ dex and exceed even the linear worst case.

4. DISCUSSION

The conclusion is falsifiable on two distinct axes that must not be conflated. **(i) Abundance vs TNG**

Table 1. Itemized stellar-mass systematic budget for $z \approx 4$–6 massive JWST galaxies, replacing the single "~ 1 dex" figure. Each axis drives a downward revision of $M_\star$. Combining axes in quadrature (independent) gives the realistic committed budget of 0.55 dex; 0.46 dex excluding the contested top-heavy-IMF term (#1); 1.30 dex in the fully correlated (linear-sum) worst case.

#	Source of $M_\star$ systematic	central (dex)	plausible range	grounding
1	IMF choice (Chabrier $\to$ top-heavy at high z)	0.30	0.1–1.0	Lapi et al. (2024); Steinhardt et al. (2
2	SFH prior / outshining (parametric vs nonparametric)	0.30	0.2–0.5	Harvey et al. (2025)
3	SPS model + nebular continuum (BC03 vs BPASS)	0.20	0.1–0.3	Choe et al. (2026); Cochrane et al. (2
4	Dust–age–metallicity degeneracy	0.15	0.1–0.25	Choe et al. (2026)
5	AGN / "Little Red Dot" host contamination (pop.-averaged)	0.20	0.1–1.0*	Zhuang et al. (2026); Kocevski et al.
6	Eddington bias (steep MF $\times$ mass-error convolution)	0.15	0.1–0.25	Adams et al. (2023); Grazian et al. (2

*Per-object LRD contamination can reach orders of magnitude, but only a fraction of the massive sample are LRDs, so the population-averaged budget is ≈ 0.2 dex.

Table 2. Downward stellar-mass shift $\Delta \log M_\star$ (dex) required to erase the observed massive-galaxy abundance excess over IllustrisTNG, across the observed excess factor and the massive-end simulated SMF slope $s = d \log n / d \log M_\star$. Entries ≤ 0.55 dex lie within the committed quadrature budget; the $z \simeq 5$ row uses the aperture-corrected total-mass excess.

Observed excess	$s{=}{-}1.4$	$s{=}{-}1.6$	$s{=}{-}1.8$	$s{=}{-}2.0$
2× (conservative)	0.22	0.19	0.17	0.15
2.04× ($z{\simeq}5$)	0.22	0.19	0.17	0.15
13.6× ($z{\simeq}7$–9)	0.81	0.71	0.63	0.57
20× (extreme)	0.93	0.81	0.72	0.65

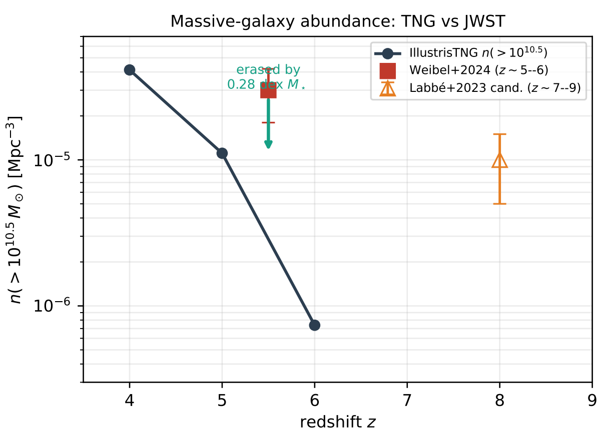

Figure 1. Cumulative number density $n(> 10^{10.5}\,M_\odot)$ vs redshift. IllustrisTNG (dark line/circles) compared to observed anchors (Weibel+2024 at $z \simeq 5$; Labbé+2023 candidates at $z \simeq 7$–9). The $z \simeq 5$ excess over TNG (on the total-mass footing) is erased by a 0.20 dex downward stellar-mass shift, well within the 0.46–0.55 dex committed systematic budget (green arrow).

(from Table 2). The $z \simeq 5$ null reverts to a tension only if the true mass-systematic budget is < 0.20 dex; the $z \simeq 7$–9 point reverts below its slope-dependent threshold of ≈ 0.72 dex (at $s = -1.58$). The committed 0.46–0.55 dex budget clears the first with a factor ~ 2 of margin but falls *short* of the second — hence $z \simeq 7$–9 is labelled marginal. **(ii) ΛCDM physical feasibility (the hard bound).** Abundance-matching the observed $n = 3 \times 10^{-5}$ Mpc^{-3} at $z = 5$ to a self-contained Sheth–Tormen halo mass function (Planck $f_b = 0.157$) gives $M_{\rm halo} = 1.0 \times 10^{12}\,M_\odot$ (log = 12.00; HMF sanity $n(> 10^{12}, z{=}5) = 3.0 \times 10^{-5}$ Mpc^{-3}, self-consistent). The unshifted $M_\star = 10^{10.50}$ then implies a baryon-conversion efficiency $\epsilon = M_\star/(f_b\,M_{\rm halo}) = 0.20$ — precisely the fiducial ΛCDM value and well under the $\epsilon \leq 1$ hard ceiling (Boylan-Kolchin 2023); after the 0.20 dex shift, $\epsilon = 0.13$. The $\epsilon = 1$ ceiling is breached only if the true masses are +0.70 dex *higher* than reported — a change opposite in sign, and 2.5× larger, than any plausible downward systematic. The observed abundance is thus nowhere near physically impossible in ΛCDM. This benchmark is robust to the HMF prescription: across Sheth–Tormen, Tinker-2008 and Press–Schechter and across halo mass-definition choices, the abundance-matched $M_{\rm halo}$ moves by $\lesssim 0.12$ dex, shifting ϵ by $\leq\sim 0.06$ (a Tinker-2008 $200m$ mass function gives $\log M_{\rm halo} = 11.90$, $\epsilon \approx 0.26$) and the +0.70 dex breach threshold by $\leq\sim 0.11$ dex (to $\approx +0.59$ dex). The direction of this HMF sensitivity runs mildly *against* consistency (higher ϵ, thinner margin), but $\epsilon \approx 0.2$–0.26 is nowhere near unity and the verdict is unchanged.

Because the unshifted abundance already implies $\epsilon \approx 0.20$ — the fiducial ΛCDM star-formation efficiency — the $z \simeq 5$ offset is a *TNG feedback/SMF-calibration* tension, not a ΛCDM stress test; the 0.20 dex shift only reconciles TNG's specific SMF. The $z \simeq 7$–9 excess, while it may sit within the fully correlated worst-case budget, both exceeds the committed quadrature budget and rests on photometric candidate masses that spectroscopy has repeatedly revised downward. A genuine, distinct residual remains: the spectroscopically confirmed compact quiescent galaxies at $z > 6$, whose inferred number density lies ~ 2 dex above baseline predictions and is not dissolved by the star-forming systematics above; this is the one ΛCDM-relevant residual, warrants dedicated follow-up, and is not claimed to be resolved here.

5. CONCLUSION

The "too massive, too early" reading of JWST massive-galaxy counts at $z \simeq 5$ is not a ΛCDM stress test. The *unshifted* observed abundance already sits at a baryon-conversion efficiency $\epsilon \approx 0.20$ — the fiducial ΛCDM star-formation efficiency — so ΛCDM is comfortably satisfied with no mass revision at all; reaching the $\epsilon = 1$ physical ceiling would require masses $\approx +0.70$ dex *higher* than reported, not lower. What the $z \simeq 5$ offset actually probes is TNG's *specific* feedback and SMF calibration: once TNG is placed on the observed total-mass footing (an aperture correction of $+0.13$ dex), the factor-2.04 excess is erased by a 0.20 dex downward stellar-mass shift, $\sim 0.4\times$ the committed 0.46–0.55 dex systematic budget and covered even without a top-heavy IMF, so we find no robust tension with TNG at $z \simeq 5$ — subject to the ± 0.10 dex Poisson floor of the 15–20-object simulated anchor and to the caveat that an effective comparison redshift of $z \simeq 5.5$ would raise the required shift to ≈ 0.42 dex (still within budget, thinner margin). We do not claim a measurement of consistency, only that the data at these redshifts neither require a departure from ΛCDM nor a change to TNG beyond its known stellar-mass systematics. The apparent $z \simeq 7$–9 excess is a weaker case: it requires ≈ 0.72 dex, which exceeds the committed budget, and rests on unconfirmed photometric masses — we label it outside the realistic budget and marginal, alongside the one genuine ΛCDM-relevant residual, the spectroscopically confirmed quiescent galaxies at $z > 6$ whose ~ 2 dex excess is not dissolved by these systematics and is not claimed to be resolved here. This is a descriptive, automated confrontation; it has not been validated by human review.

REFERENCES

Adams, N. J., et al. 2023, MNRAS, 518, 4755

Boylan-Kolchin, M. 2023, NatAs, 7, 731

Choe, J., et al. 2026, A&A, 707, A29

Cochrane, R. K., et al. 2025, ApJL, 978, L42

Grazian, A., et al. 2015, A&A, 575, A96

Harvey, T., et al. 2025, ApJ, 978, 89

Kocevski, D. D., et al. 2025, ApJ, 986, 126

Labbé, I., et al. 2023, Nature, 616, 266

Lapi, A., et al. 2024, Universe, 10, 141

Nelson, D., et al. 2019, ComAC, 6, 2

Steinhardt, C. L., et al. 2023, ApJL, 951, L40

Weibel, A., et al. 2024, MNRAS, 533, 1808

Zhuang, M.-Y., et al. 2026, ApJ, 999, 31